

Adhish Aravind

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

Software Engineer with production experience building AI-enabled systems and retrieval-augmented workflows. Strong ownership of LLM integration, prompt orchestration, and asynchronous processing with proven ability to integrate AI capabilities into reliable backend systems while delivering measurable business impact.

Skills

AI & Machine Learning: LLM Integration, Retrieval-Augmented Generation (RAG), Prompt Engineering, OpenAI API, NLP, LangGraph, LangChain, Agentic Workflows, Multimodal Pipelines

Backend & Frameworks: Python, Django, Django REST Framework, FastAPI, REST APIs, WebSockets, PostgreSQL, MongoDB

Async & Infrastructure: Celery, Redis, RabbitMQ, Event-driven Architecture, State Machines, Docker, AWS, GCP, CI/CD, Nginx

Core Concepts: Vector Databases, Semantic Search, Rate Limiting, Idempotency, JWT Auth, Testing

Experience

AI Engineer

Aug 2025 – Present

Emirion Techlab

Remote

- Built and maintain a RAG-based customer support chatbot (FastAPI) deployed as an embedded widget on client websites for FAQ and product support
- Fixed retrieval failures on tabular/numeric queries by reworking chunking and chunk-linkage, addressing RAG's loss of row-column relationships needed for exact figures
- Reduced response latency by implementing caching for embeddings and repeated query-response pairs based on historical usage patterns
- Containerized the service with Docker and managed cloud deployment for the chatbot backend
- Designed and exposed FastAPI endpoints to serve AI/RAG services to client-facing applications
- Added client-facing controls (temperature tuning, knowledge base updates via admin panel) enabling clients to adjust chatbot behavior without engineering support
- Currently scoping ERP integration to enable natural-language queries over sales performance data (targets, blockers, team performance)

Software Engineer — Contract

Nov 2025 – Mar 2026

Guiding Dreams

Remote

- Built and deployed AI-powered WhatsApp education platform end-to-end using Django, PostgreSQL, OpenAI API, and Celery, integrating LLM-based content generation and contextual responses via Twilio API.
- Implemented stateful retrieval-augmented generation (RAG) system with Celery Beat automated week-wise progression to dynamically serve lesson content and transcripts, decreasing teacher workload by 40%.
- Architected multi-conversation AI engine with prompt orchestration supporting 5 interaction types, maintaining conversation history and state across concurrent user sessions.
- Containerized application with Docker, established GitHub Actions CI/CD to GCP with Nginx reverse proxy; deployed async pipelines using Celery workers and Redis for background AI inference, reducing API latency by 50%.

Data Scientist Intern (Applied ML Systems)

Mar 2024 – Dec 2024

AIvariant

Bengaluru

- Built end-to-end NLP pipelines for resume classification and recommendation systems using machine learning, including TF-IDF vectorization, tokenization, and feature engineering for large datasets.
- Developed collaborative filtering and content-based recommendation models, deployed real-time ML inference systems using Streamlit with optimized latency.

Projects

Sigint AI — Operational Intelligence & Incident Correlation Platform

[GitHub](#) | [Live Demo](#)

FastAPI, React, PostgreSQL, MongoDB, RabbitMQ, OpenAI API, Docker

- Architected a 5-stage async ingestion pipeline (RabbitMQ, circuit breaker, exponential backoff retries) aggregating cross-platform signals from Slack, Jira, GitHub, and Gmail into PostgreSQL and MongoDB.
- Designed a hybrid correlation engine combining entity extraction, OpenAI embedding cosine similarity, keyword overlap, and time decay to auto-correlate telemetry into incidents, eliminating false groupings.
- Built LLM signal classification (GPT-4o-mini) and multi-agent root-cause analysis (GPT-4o) with anti-hallucination safeguards, streaming diagnostic progress via SSE; deployed as 9 microservices on Oracle Cloud behind Nginx.

Agentic LLM Content Orchestration System

[GitHub](#)

LangGraph, LangChain, OpenAI/Gemini APIs, RAG, Async Pipelines

- Designed production-grade agentic system using LangGraph with deterministic state transitions to orchestrate multi-step content workflows across planning, routing, execution, and reduction stages.

- Built intelligent router selecting closed-book, hybrid, or RAG modes with citation enforcement, structured outputs, and fan-out worker architecture for parallel section generation and multimodal creation.
- Evolve — AI-Based Nutrition Coach** [GitHub](#)
- Django REST, React, PostgreSQL, WebSockets, Celery, Gemini API, Docker, GCP*
- Built AI-powered nutrition coaching platform with real-time chat using WebSockets and personalized meal planning powered by Gemini API with context-aware prompt engineering.
 - Implemented JWT authentication, onboarding flows, and Celery-based async processing for concurrent Gemini API interactions.

Education

College of Engineering Thalassery	Kannur, India
<i>Bachelor of Technology in Information Technology — First Class</i>	<i>2023</i>